

Robust Learning Systems under Distribution Shift

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Abstract

We present an empirical study of robust inference in latent representation models. The proposed regularization method constrains the model while preserving accuracy. An ablation analysis measures the contribution of every component.

1. Introduction

Reliable systems must remain robust when observations differ from training data. This requirement is especially important in scientific and safety-critical settings. Our empirical evaluation covers uncertainty, calibration, and generalization.

2. Method

We learn a latent space and constrain its geometry using a structured objective. Regularization reduces variance without suppressing meaningful representations. Inference is performed with a lightweight approximation at test time.

3. Results

The robust model improves mean accuracy from 81.4 percent to 87.9 percent. Ablation results indicate that both constraints contribute to the final score. These findings support the proposed approach across all evaluated datasets.